INTRODUCTION TO DATA SCIENCE

GROUP PROJECT

Data Science for Better Service Delivery in South Africa

From a real public problem to a statistically validated machine-learning solution

Assessment type	Group project
Group size	Maximum 6 students
Total marks	100
Weighting	As prescribed in the module assessment plan
Due date / presentation	Presentations Starts from 5th October -17th October 2026

CENTRAL CHALLENGE

How can responsible data science help a South African public-service stakeholder identify need, prioritise limited resources, or improve the reliability and reach of a service?

Student names • student numbers • group number • lecturer • submission date

1. PROJECT PURPOSE

In a team of no more than six, identify a real service-delivery problem in South Africa and develop an end-to-end, reproducible machine-learning solution. The project must move beyond model accuracy: it must establish a credible problem, use lawful and relevant data, compare robust baselines with neural models, control overfitting, optimise one justified champion, test which components matter, and determine whether the observed improvement is statistically credible.

Learning outcomes

- Frame a public-service problem as a measurable data-science task with clear stakeholders and success criteria.
- Acquire, audit, visualise and prepare real data without leakage.
- Build reproducible traditional and neural-learning pipelines and evaluate them fairly.
- Diagnose overfitting, optimise a selected model, conduct an ablation study and apply a paired statistical test.
- Deploy the selected model through a usable user interface that supports stakeholder-facing prediction and responsible interpretation.
- Communicate limitations, ethical risks and actionable recommendations for a South African context.

2. PROJECT SCOPE AND DATA

Select a classification problem connected to service delivery. A regression or forecasting problem requires prior lecturer approval and a statistically appropriate paired alternative to McNemar's test. The outcome must be meaningful to a public-service stakeholder—not merely convenient to predict.

Possible direction	Example prediction target	Potential public value
Basic-service vulnerability	Classify municipalities, wards or areas into service-need/risk categories	Support equitable prioritisation of water, sanitation, electricity, refusing removal or housing interventions
Faults and citizen reports	Classify complaint type, urgency, escalation or likely resolution status	Improve triage and response planning

Possible direction	Example prediction target	Potential public value
Infrastructure reliability	Classify outage, failure or interruption risk	Support preventive maintenance and resource allocation
Access and inclusion	Classify areas or households at risk of inadequate service access	Identify gaps and monitor equitable reach

Approved source route

Use a relevant dataset from Zindi or another authoritative, lecturer-approved platform. The source must provide sufficient metadata, lawful reuse conditions and a defensible connection to South African service delivery.

Source	How it may be used	Link
Statistics South Africa	Census and service-delivery indicators covering housing, water, sanitation, electricity and solid waste at provincial/local-municipality level.	https://www.statssa.gov.za/?p=17836
Zindi	Select an open challenge or dataset whose target can be reframed responsibly for a South African service-delivery question.	https://zindi.africa/competitions
Another approved platform	Municipal open-data portals, national departments, research repositories or open geospatial sources; obtain written lecturer approval before modelling.	Record the exact dataset URL and licence in the report.

Platform competition rules are separate from this course. Your course group may contain up to six members, but any optional external competition entry must comply with that platform's own team rules.

Data approval checkpoint

Submit before modelling	Required evidence

Submit before modelling	Required evidence
One-page concept notes	Problem, stakeholder, prediction unit, target, intended decision and anticipated public value.
Dataset card	Source URL, owner, licence/terms, collection period, unit of analysis, features, target, size, missingness, sensitive attributes and known limitations.
Feasibility check	A usable target, adequate sample size/classes, no prohibited personal data, computing suitable for the module, and an evaluation plan.

3. ASSESSMENT QUESTIONS

Answer every question in the submitted technical report. Marks in brackets indicate the relative depth expected.

QUESTION 1 **Total Marks [10]**

1.1 Define the South African service-delivery problem, affected population, decision-maker and intended benefit. Convert it into a classification task by specifying the unit of prediction, inputs, target classes and prediction horizon. (4)

1.2 State one primary technical metric and at least one service-delivery success indicator. Explain the cost of false positives and false negatives. (2)

1.3 Justify the dataset's relevance, provenance, licence, coverage and limitations. Identify POPIA, privacy, fairness and misuse risks and propose safeguards. (4)

QUESTION 2 **Total Marks [18]**

2.1 Create a data dictionary and audit class balance, duplicates, missingness, invalid values, outliers and possible leakage. (4)

2.2 Present at least five purposeful visualisations: target distribution; missingness/data quality; two feature–target relationships; and one geographic, temporal or subgroup view. Each figure must have a title, units, readable labels and an interpretation. (6)

2.3 Split data before learning transformations. Justify train/validation/test or nested cross-validation design, including stratification/group/time constraints. Reserve the test set for final evaluation only. (3)

2.4 Implement cleaning, encoding, scaling, imputation, feature engineering and imbalance treatment inside a reproducible pipeline. Explain choices and prevent leakage. (5)

QUESTION 3 **Total Marks [18]**

3.1 Build a transparent naïve baseline (for example, majority/stratified DummyClassifier) and explain what it establishes. (2)

3.2 Train at least four substantive models on identical data splits: at least two traditional models and at least two neural-learning models. The naïve baseline does not count among the four. (10)

3.3 Traditional examples include Logistic Regression, Decision Tree/Random Forest, Support Vector Machine, Naïve Bayes or gradient boosting. Neural requirements include an MLP and one task-appropriate second architecture (for example, CNN for imagery, GRU/LSTM for sequences, or an approved tabular neural network). (3)

3.4 Document architecture/hyperparameters, loss, optimiser, epochs, batch size, regularisation, early stopping, seeds, compute environment and reproducible training procedure. (5)

3.5 Provide a concise rationale linking each model family to the data and problem. (2)

QUESTION 4 **Total Marks [16]**

4.1 For every substantive model, report the same held-out metrics. Include accuracy and macro-F1; add precision, recall, weighted-F1 and ROC-AUC as appropriate. For imbalance, include PR-AUC or justify its omission. (4)

4.2 Provide a classification report and confusion matrix for every model; provide ROC curves on one common figure (one-vs-rest macro/micro curves for multiclass). (5)

4.3 Compare all models with a clear table and bar chart. Show cross-validation means and variability or confidence intervals where feasible. Select the provisional champion using validation—not test—performance. (3)

4.4 Test for overfitting using train-versus-validation performance and learning curves; for neural models also show training/validation loss and metric curves. Diagnose gaps and discuss early stopping, dropout, weight decay, pruning or other controls used. (4)

QUESTION 5 **Total Marks [10]**

5.1 Optimize only the strongest validation model using a documented method such as grid search, random search, Bayesian optimisation or Optuna. Define the search space and scoring rule. (4)

5.2 Use cross-validation or a strict validation set; do not tune on the held-out test set. Compare pre- and post-optimisation results and report the chosen hyperparameters. (4)

5.3 Evaluate the final locked pipeline once on the test set and explain whether the improvement is practically meaningful. (2)

QUESTION 6 **Total Marks [10]**

6.1 Conduct at least three controlled ablations. Remove or alter one feature group, preprocessing step, imbalance method, regulariser or architecture component at a time while keeping split, seed and evaluation protocol fixed. Report on the change in performance and interpret what the system depends on. (5)

6.2 Compare the optimized champion with the strongest comparator on the same held-out cases using McNemar's test. Show the 2×2 correctness table, state H_0/H_1 and $\alpha = 0.05$, report the statistic/exact p-value, and distinguish statistical from practical significance. Use the exact binomial version when discordant counts are small. (5)

QUESTION 7 **Total Marks [10]**

7.1 Deploy the final optimized model through a working user interface. The interface may be developed with Gradio, Streamlit, Flask, FastAPI with a web front end, or another lecturer-approved framework. A notebook cell alone is not a user interface. (2)

7.2 Allow a user to enter or upload valid input, run the complete saved preprocessing-and-model pipeline, and display the predicted class, confidence/probability where appropriate, and a brief plain-language explanation. (2)

7.3 Include input validation, helpful error messages, class/feature guidance, a limitation or responsible-use notice, and no exposed credentials or personal data. (1)

7.4 Provide deployment instructions and either a stable demonstration link or a locally reproducible launch command. Demonstrate at least three valid cases and one invalid-input case. (2)

QUESTION 8 **Total Marks [7]**

8.1 Translate findings into a realistic recommendation for the stakeholder. State limitations, uncertainty, fairness findings, monitoring needs and conditions under which the model should not be used. (2)

8.2 Submit a reproducible repository/package with README, environment/requirements, fixed seeds, execution order, source links, data dictionary, saved model/pipeline and user-interface code. Do not redistribute data if its licence forbids it. (3)

8.3 Deliver a concise team presentation and live or recorded interface demonstration. Defend the modelling choices and include a signed contribution statement; individual marks may be moderated where evidence of contribution differs. (2)

4. REQUIRED EVIDENCE AND EXPERIMENTAL RULES

Requirement	Minimum acceptable evidence
Data split integrity	A diagram or written sequence showing split-before-fit; preprocessing learned on training folds only; shared held-out cases for every final comparison.
Model set	Naïve baseline + ≥4 substantive models: ≥2 traditional and ≥2 neural. Hyperparameters and random seeds disclosed.
Core visual evidence	EDA figures, model comparison chart, confusion matrices, ROC curves, learning curves, neural loss/metric curves and ablation plot/table.
User-interface deployment	Working interface loads the saved final pipeline, accepts valid input, returns a prediction and confidence where appropriate, handles invalid input, communicates limitations, and can be launched from documented instructions.
Classification evidence	Per-model classification reports with class labels and support; class-sensitive interpretation rather than accuracy alone.
Optimisation	Search strategy, ranges, validation score, selected configuration and locked final test result.
Ablation	≥3 one-change-at-a-time experiments, same evaluation protocol, performance deltas and interpretation.
McNemar validation	Paired predictions on identical test cases; table counts: both correct, champion only correct, comparator only correct, both wrong; hypotheses, α, p-value and conclusion.
Overfitting	Quantified train–validation gap plus learning curves; mitigation and limitations discussed.

Requirement	Minimum acceptable evidence
Reproducibility	Code runs from documented instructions; no hidden manual steps, secrets or local-only paths.

Submission package

Item	Format / limit
Technical report	PDF, suggested 15–20 pages excluding cover, references and appendices.
Code	Clean notebook(s) and/or scripts plus README and requirements/environment file.
Deployed user interface	Interface source code, saved final pipeline/model, launch instructions, dependency file, and stable demonstration link where available.
Evidence bundles	Figures, metric tables, predictions used in McNemar's test, model/experiment log and dataset card.
Presentation	Suggested 10–12 minutes plus questions; every member must participate.
Contribution statement	One page signed by all members, including roles and specific outputs.

Academic and responsible-use conditions

- Reference every dataset, library, external code fragment and source. Clearly label adapted code and AI-assisted work according to institutional policy.
- Do not upload personal, confidential or access-controlled records to public repositories or external tools.
- A high score requires correct reasoning and reproducible evidence; a complex model without defensible evaluation will not earn automatic credit.
- Fabricated data provenance, results or contribution claims may be handled under the institution's academic-integrity procedures.

5. MARKING RUBRIC

Award marks within each criterion according to the descriptor that best matches the submitted evidence. Intermediate marks may be used. A criterion with no verifiable evidence receives zero.

Criterion	Marks	Excellent 80–100%	Competent 65–79%	Developing 50–64%	Limited 1–49%	Absent 0%
1. Problem, stakeholder, data & ethics	**10**	Compelling service-delivery needs; precise task and costs; authoritative, well-licensed dataset; excellent contextual, POPIA and fairness reasoning.	Clear and relevant formulation; suitable data and good risk discussion; minor gaps.	Partly defined task or weak public-value link; limited data/ethics justification.	Unclear or unsuitable problem/data; major ethical or provenance gaps.	No credible evidence.
2. Data audit, EDA, visualisation & pipeline	**18**	Thorough audit; insightful, accessible visuals; rigorous split; leakage-safe, reproducible preparation with strong justification.	Sound audit and visuals; mostly correct pipeline; small omissions.	Basic EDA/preparation; weak interpretations or some pipeline risks.	Minimal/incorrect preparation; serious leakage or unusable visuals.	No credible evidence.

Criterion	Marks	Excellent 80–100%	Competent 65–79%	Developing 50–64%	Limited 1–49%	Absent 0%
3. Baseline & ≥4 models	**22**	Naïve baseline plus compliant, reproducible 2+2 model set; strong rationales and complete neural/training detail.	All model requirements met and mostly reproducible; minor documentation gaps.	Four models attempted but a family/architecture, rationale or reproducibility requirement is weak.	Insufficient models or largely invalid/unreproducible training.	No credible evidence.
4. Evaluation, comparison & overfitting	**16**	Consistent multi-metric evaluation; complete reports/matrices/ ROC; excellent visual comparison and quantified overfitting diagnosis/controls.	Correct core evidence and sensible interpretation; minor gaps.	Partial metrics/plots or superficial overfitting analysis.	Inconsistent evaluation, test misuse, or major required evidence missing.	No credible evidence.

Criterion	Marks	Excellent 80–100%	Competent 65–79%	Developing 50–64%	Limited 1–49%	Absent 0%
5. Champion optimisation	**10**	Validation-led, well-designed search; reproducible configuration; honest pre/post and one-time test evaluation with practical interpretation.	Correct optimisation with good evidence; limited search/interpretation detail.	Optimisation attempted but weak protocol or incomplete comparison.	Test-set tuning, unjustified search, or non-reproducible claim.	No credible evidence.
6. Ablation & McNemar validation	**10**	≥3 controlled ablations with insightful deltas; correct paired test, assumptions, contingency table, p-value and nuanced conclusion.	Correct minimum ablations and McNemar test; small interpretive gaps.	Incomplete controls or minor statistical errors, but approach is recognisable.	Ablation absent/invalid or statistical test fundamentally misapplied.	No credible evidence.

Criterion	Marks	Excellent 80–100%	Competent 65–79%	Developing 50–64%	Limited 1–49%	Absent 0%
7. User-interface deployment	7	Working, polished interface runs the saved end-to-end pipeline; validates input; presents useful predictions and confidence; includes safeguards and fully reproducible deployment.	Functional interface and documented launch; minor usability, explanation or validation gaps.	Basic interface works for limited cases but deployment, validation or documentation is incomplete.	Interface is largely non-functional, bypasses the final pipeline or cannot be reproduced.	No credible evidence.
8. Conclusions, reproducibility & team defence	7	Actionable, cautious recommendation; transparent limitations/fairness; complete code and deployment reproducibility; polished defence and credible shared contribution.	Clear conclusion and usable package/presentation; minor omissions.	Generic conclusion, partial reproducibility or uneven defence.	Unsupported claims, unusable package or weak contribution evidence.	No credible evidence.

TOTAL: 100 MARKS

Suggested team roles

Roles may overlap, but accountability must be visible: project/stakeholder lead; data governance and preparation lead; EDA/visualisation lead; traditional modelling lead; neural modelling lead; and evaluation, deployment and reproducibility lead. All members remain responsible for understanding and defending the complete pipeline and user interface.